# Yelp Integration

## 1. Overview

### 1.1 Description

**Yelp Integration** connects **Yelp** (business directory and lead generation platform) with **Newo Platform**.

It enables:

- Automatic processing of incoming Yelp leads (job requests)
- AI-powered responses to customer inquiries via Yelp chat
- Human manager takeover with configurable auto-resumption
- Message deduplication to prevent duplicate agent responses
- Developer/test mode for safe development without contacting customers
- Custom per-business conversation scenarios

This integration is designed for **service businesses and contractors on Yelp** who need **automated lead qualification and customer communication through a conversational AI agent** (chat channel).

### 1.2 Use Cases

- When a new Yelp lead arrives → AI analyzes lead details, identifies missing info, and asks clarifying questions
- When a customer sends a follow-up message → AI continues the conversation, maintaining context
- When a manager joins the conversation → Automation pauses, then resumes after a configurable delay
- When the manager wants to take full control → Permanently pause automation via human intercept flag
- When developing/testing → Dev mode processes events but doesn't send responses to real customers

### 1.3 Supported Features

| Feature | Supported | Notes |
|---|---|---|
| Lead Processing | ✅ | Extracts job name, email, postal, survey answers |
| Customer Message Handling | ✅ | Full conversation context, deduplication |
| AI Lead Response | ✅ | LLM-powered analysis and clarifying questions |
| Manager Takeover | ✅ | Auto-pause on manager join, timer-based auto-resume |
| Human Intercept | ✅ | Permanent pause flag for manual control |
| Message Deduplication | ✅ | Hamming distance similarity (threshold 0.9) |
| Developer Mode | ✅ | Processes events without sending to Yelp |
| Custom Scenarios | ✅ | Per-business behavior customization |
| Message Delivery via Zapier | ✅ | Outbound messages routed through Zapier webhook |
| Session Cleanup | ✅ | Auto-clear persona attributes on session end |

| | | |
|---|---|---|
| Phone Lead Handling | ❌ | Phone type events are received but not processed |
| Review Management | ❌ | No review reading or response capabilities |
| Yelp Ads Integration | ❌ | Not implemented |

## 2. Requirements

Before installation:

- Active **Yelp for Business** account with lead generation enabled
- **Zapier** account with a webhook configured to relay messages to Yelp
- Webhook endpoint for receiving Yelp events (configured in Yelp → Newo)

## 3. How to Setup

### 3.1 Step 1 — Configure Zapier Webhook

1. Create a Zapier Zap that receives webhook POSTs and sends messages to Yelp
2. Copy the Zapier webhook URL (this is how the agent sends responses to Yelp)
3. Set the URL in the `yelp_zapier_webhook` attribute

### 3.2 Step 2 — Connect in Platform

1. Open **Newo → Projects**
2. Set the following attributes in **Builder / Attributes**:

| Attribute | Required | Description |
|---|---|---|
| `yelp_zapier_webhook` | ✅ | Zapier webhook URL for sending messages to Yelp |
| `yelp_integration_enabled` | ❌ | Master switch (default: `True`) |
| `yelp_take_over_enabled` | ❌ | Auto-pause/resume on manager join (default: `True`) |
| `yelp_take_over_delay` | ❌ | Seconds before auto-resume after manager leaves (default: `120`) |
| `yelp_human_intercept` | ❌ | Permanently pause automation (default: `False`) |
| `yelp_dev_mode` | ❌ | Developer mode — process but don't send (default: `True`) |
| `yelp_custom_scenario` | ❌ | Custom instructions/scenario for agent behavior |

3. Click **Save + Publish All**
4. On publish, the SetupFlow automatically:
    - Creates HTTP connector for Zapier communication
    - Creates timer connector for takeover delay management
    - Initializes all attributes with defaults

> ***Important:*** *`yelp_dev_mode` defaults to `True`. Set to `False` when ready for production to enable actual message sending.*

## 4. How to Use

This section explains how the integration works, how to configure automation, and how to test it.

## 4.1 How the Integration Works

The integration receives Yelp events via webhook and routes them based on type: `lead` (new inquiry), `user_message` (customer reply), `business_message` (manager reply), or `phone` (ignored). Outbound messages are sent to Yelp through a Zapier webhook relay.

- A **Trigger** is a system event that starts a flow
- An **Action** is the operation performed in response

**Available Triggers**

| Trigger | Event | Description |
|---|---|---|
| **Yelp Event** | `yelp_income` | Incoming webhook from Yelp (lead, message, etc.) |
| **Send Reply** | `send_external_reply` | When the agent generates a response |
| **Session Ended** | `session_ended` | Conversation cleanup |
| **Takeover Timer** | `timer` | Auto-resume after manager leaves |
| **Setup** | `project_publish_finish` | Initializes integration on deploy |

**Available Actions**

- **Process Lead** — Extract lead details, inject into prompt, generate AI response
- **Send Message** — POST to Zapier webhook with `{lead_id, message}`
- **Start Takeover** — Pause automation, start timer for auto-resume
- **Resume Automation** — Clear takeover flag, optionally respond to pending questions
- **Cleanup Session** — Clear all persona attributes on session end

## 4.2 Architecture Diagrams

**Overall Sequence**

```
sequenceDiagram
    participant Y as Yelp
    participant W as Webhook
    participant I as Yelp Integration
    participant A as ConvoAgent (LLM)
    participant Z as Zapier
    participant M as Manager

    Note over Y,Z: New Lead
    Y->>W: yelp_income (type: "lead", body: (...))
    W->>I: OnYelpEvent
    I->>I: Extract lead details (job, email, postal, survey)
    I->>I: Inject NewLeadDetails into prompt
    I->>A: urgent_message (analyze lead, ask questions)
    A-->>I: Generated response
    I->>Z: POST (lead_id, message)
    Z->>Y: Deliver to customer
```

```
    Note over Y,Z: Customer Reply
    Y->>W: yelp_income (type: "user_message")
    W->>I: OnYelpEvent
    I->>I: Deduplicate (hamming 0.9)
    I->>A: Create message act
    A-->>I: Agent response
    I->>Z: POST (lead_id, message)
    Z->>Y: Deliver to customer

    Note over M,Z: Manager Takeover
    M->>Y: Manager sends message
    Y->>W: yelp_income (type: "business_message")
    W->>I: OnYelpEvent
    I->>I: Set take_over_started = True
    I->>I: Start timer (120s default)
    Note over I: Agent paused
    I-->>I: Timer fires
    I->>I: Set take_over_started = False
    I->>A: Check if customer has pending question
    alt Pending question
        A-->>I: Response
        I->>Z: POST (lead_id, message)
    end
```

**YelpIntegrationFlow — Event Router**

```
flowchart TD
    E["yelp_income<br>webhook"] --> GATE{{"Integration<br>enabled?"}}
    GATE -->|"No"| SKIP["Skip"]
    GATE -->|"Yes"| INTERCEPT{{"Human<br>intercept?"}}
    INTERCEPT -->|"Yes"| SKIP
    INTERCEPT -->|"No"| TYPE{{"Event type?"}}
    TYPE -->|"lead"| LEAD["_onNewLeadSkill:<br>Extract details<br>Inject into
prompt<br>Generate AI response"]
    TYPE -->|"user_message"| USER["_onUserMessageSkill:<br>Deduplicate<br>Create
message act<br>Start takeover timer"]
    TYPE -->|"business_message"| BIZ["_onBusinessMessageSkill:<br>Check
duplicate<br>Set takeover flag<br>Start resume timer"]
    TYPE -->|"phone"| NOOP["No action"]
    LEAD --> SEND["SendMessageSkill"]
    USER --> AGENT["Agent generates<br>response"]
    AGENT --> SEND
    SEND --> CHECK{{"Dev mode?<br>Takeover?<br>Intercept?"}}
    CHECK -->|"Any true"| NOSEND["Skip sending"]
    CHECK -->|"All false"| POST["HTTP POST<br>to Zapier webhook<br>{lead_id,
message}"]
```

**Manager Takeover Flow**

```
flowchart TD
    BIZ["business_message<br>received"] --> FLAG["Set take_over_started<br>= True"]
    FLAG --> TIMER["Start timer<br>(yelp_take_over_delay<br>default: 120s)"]
    TIMER --> PAUSE["Agent paused:<br>SendMessageSkill<br>skips all messages"]
    PAUSE --> FIRE["Timer fires:<br>OnTakeOverTimerSkill"]
    FIRE --> CLEAR["Set take_over_started<br>= False"]
    CLEAR --> CHECK{{"Last message<br>from business?"}}
    CHECK -->|"Yes"| DONE["No action<br>(manager handled it)"]
    CHECK -->|"No"| LLM["LLM check:<br>Does customer have<br>pending question?"]
    LLM -->|"Yes"| REPLY["Generate and<br>send response"]
    LLM -->|"No"| DONE
```

**SetupFlow**

```
flowchart TD
    E["project_publish_finish<br>or import_finish"] --> SETUP["Define connectors:
<br>yelp_connector (HTTP)<br>take_over (timer)"]
    SETUP --> ATTRS["Define 7 attributes:<br>enabled, intercept,<br>takeover, delay,
<br>zapier URL, dev mode,<br>custom scenario"]
    ATTRS --> INIT["Create connectors<br>if not exist"]
    INIT --> SET["Set attributes<br>(non-destructive:<br>only if not present)"]
    SET --> DONE["Setup complete"]
```

### 4.3 Where to Test

Testing can be done in **developer mode** ( `yelp_dev_mode = True` , the default). In dev mode, the integration processes all incoming Yelp events, generates AI responses, but **does not send messages** to Yelp via Zapier. This allows safe testing of the full pipeline.

### 4.4 Testing and Verification

**To test the integration:**

1. **Setup**: Publish the project and verify connectors are created
2. **Dev Mode**: Keep `yelp_dev_mode = True` during testing
3. **Lead**: Trigger a test lead webhook — verify AI generates appropriate response with clarifying questions
4. **User Message**: Send a test customer message — verify deduplication and response generation
5. **Business Message**: Send a manager message — verify takeover activates and timer starts
6. **Timer**: Wait for takeover delay — verify automation resumes and checks for pending questions
7. **Go Live**: Set `yelp_dev_mode = False` when ready for production

**If no action occurs:**

- Ensure `yelp_integration_enabled` is `True`
- Verify `yelp_zapier_webhook` is set to a valid Zapier webhook URL
- Check `yelp_human_intercept` is `False` (unless intentionally pausing)
- Confirm `yelp_dev_mode` is `False` for production (messages won't send in dev mode)
- Verify webhook connectivity: Yelp events must reach the `yelp_income` endpoint
- Re-publish the project if attributes were changed

> *Note: This integration sends messages to real customers when dev mode is off. Always test thoroughly in dev mode before going live.*

## 5. FAQ

**Q: How does the AI respond to new leads?** A: When a lead arrives, the integration extracts all details (job name, email, postal code, survey answers) and injects them as a custom prompt section. The LLM analyzes the lead, identifies missing information, and generates a friendly response with clarifying questions (3-6 sentences).

**Q: What is the difference between human intercept and takeover?** A: **Human intercept** ( `yelp_human_intercept` ) is a permanent pause — automation stops until manually re-enabled. **Takeover** ( `yelp_take_over_enabled` ) is automatic — when a manager sends a message, automation pauses for `yelp_take_over_delay` seconds (default: 120), then resumes. Takeover is designed for managers who briefly check in; intercept is for full manual control.

**Q: Why does the integration use Zapier instead of direct Yelp API?** A: Zapier acts as a message relay middleware. This decouples Yelp API authentication from the integration, allows businesses to enrich or modify messages in Zapier flows, and provides a simpler webhook-based interface.

**Q: What is dev mode?** A: When `yelp_dev_mode = True` (the default), the integration processes all events and generates AI responses, but **does not send** them to Yelp. This allows safe development and testing. Set to `False` for production.

**Q: How does message deduplication work?** A: The integration stores the last agent message and uses hamming distance similarity (threshold 0.9) to detect near-identical messages. This prevents the agent from sending duplicate responses when the same event is processed multiple times.

**Q: Can I customize the AI's behavior per business?** A: Yes. Set the `yelp_custom_scenario` attribute with custom instructions. These are injected into the LLM prompt during lead response generation, allowing per-business tone, process, or requirement customization.

**Q: What happens when a conversation ends?** A: The `SessionEndedSkill` clears all persona attributes (last agent message, takeover flags, intercept state, message direction). This ensures a clean state for the next conversation.

**Q: Does the integration handle Yelp phone calls?** A: Phone-type events are received but not processed. The integration currently supports only chat/text-based communication through Yelp.